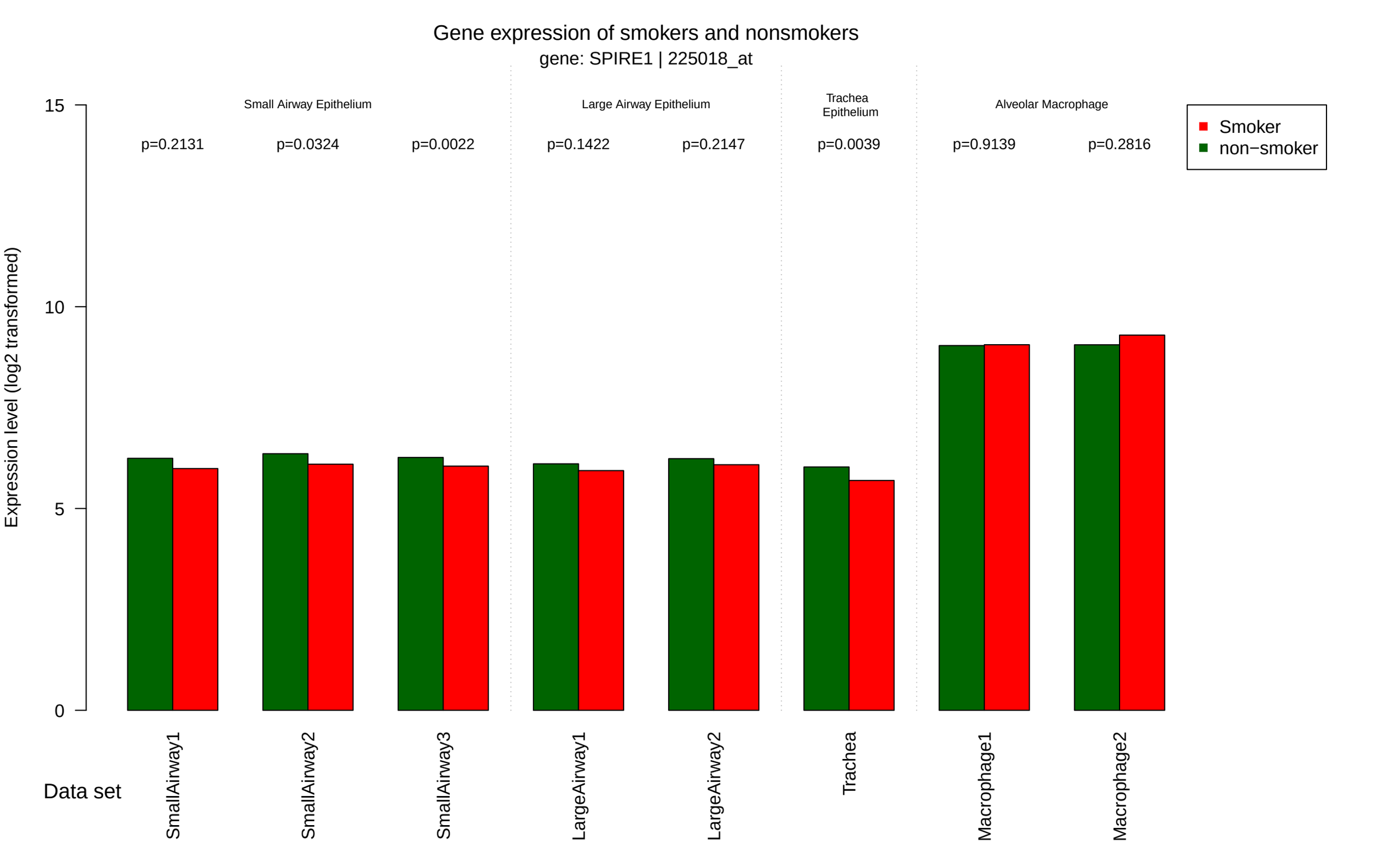
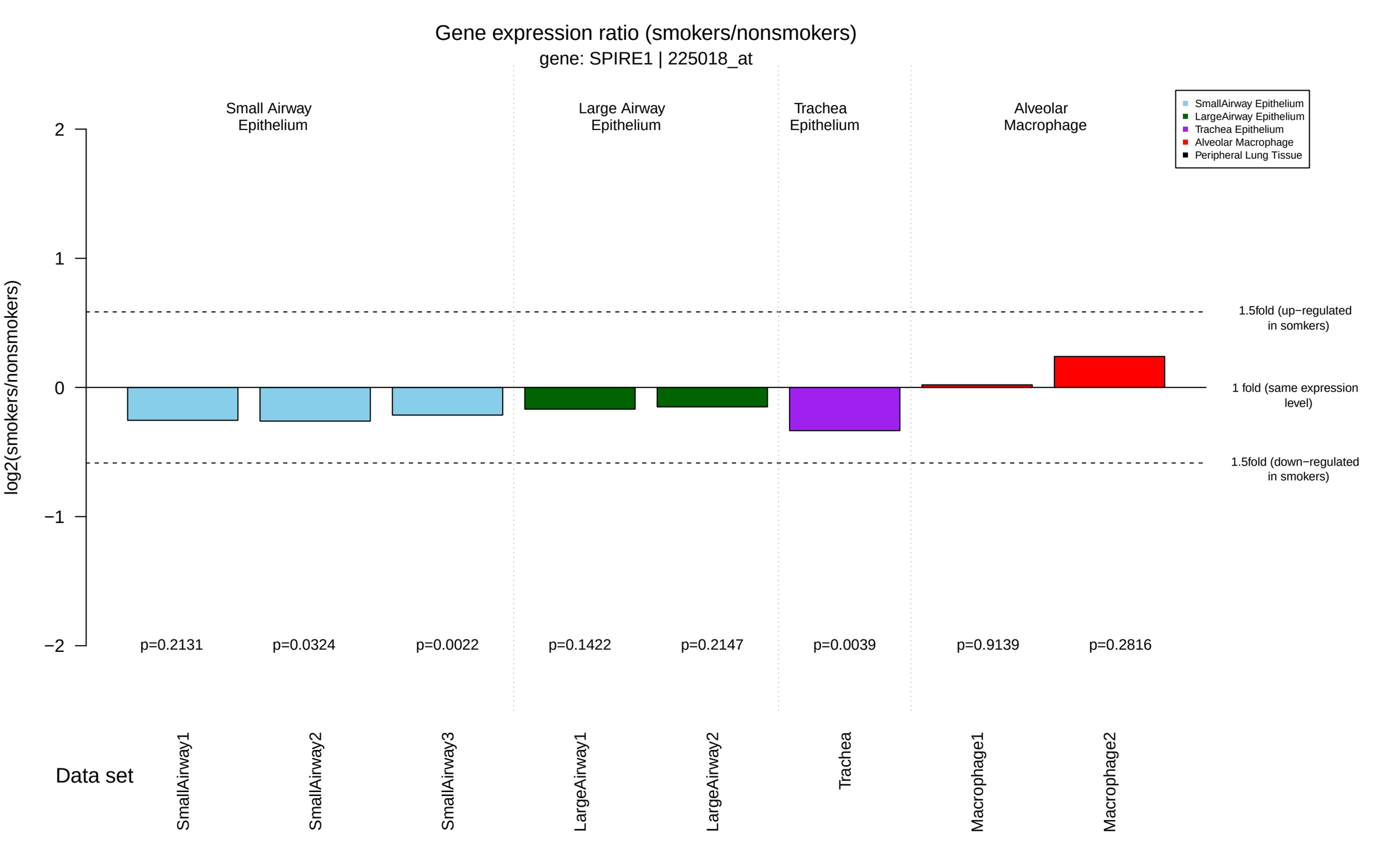
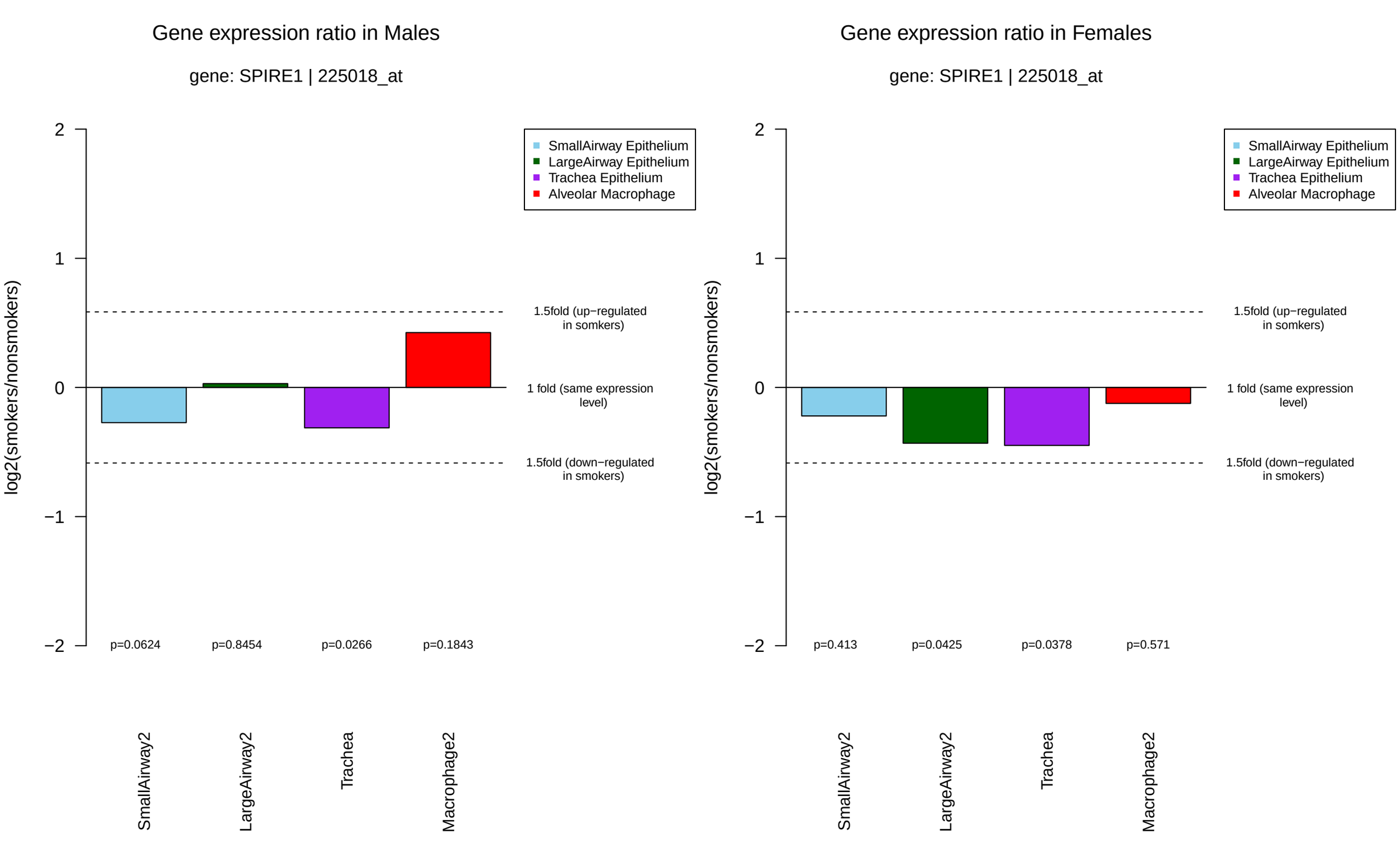
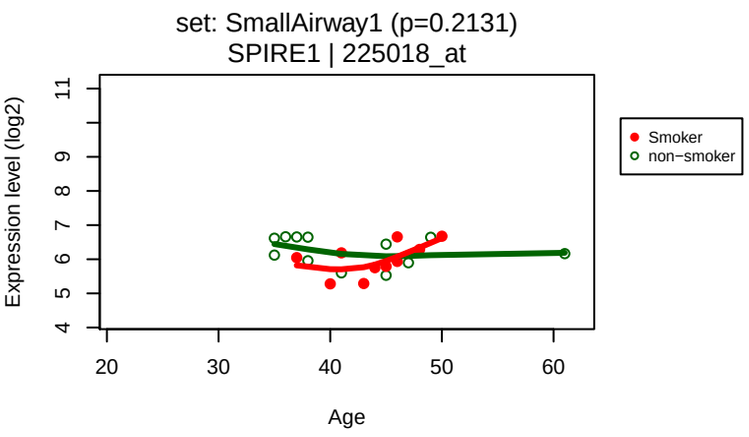
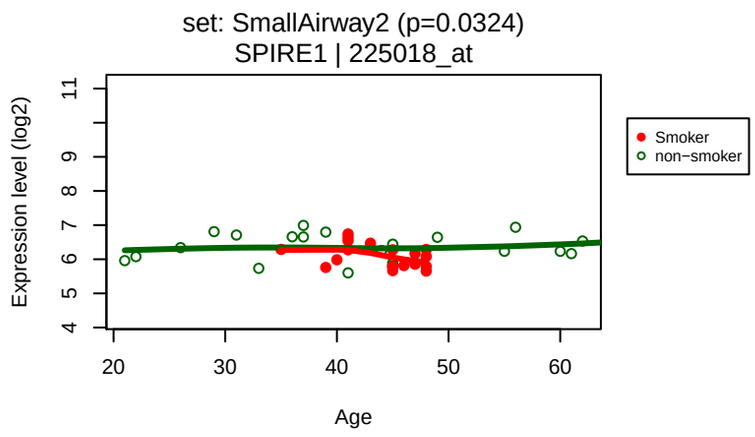
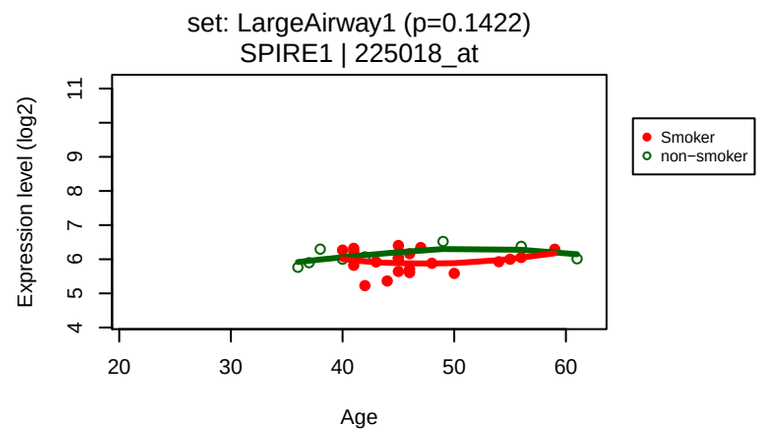

Gene expression of smokers and nonsmokers
gene: SPIRE1 | 225018_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2131
p=0.0324
p=0.0022
p=0.1422
p=0.2147
p=0.0039
p=0.9139
p=0.2816
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

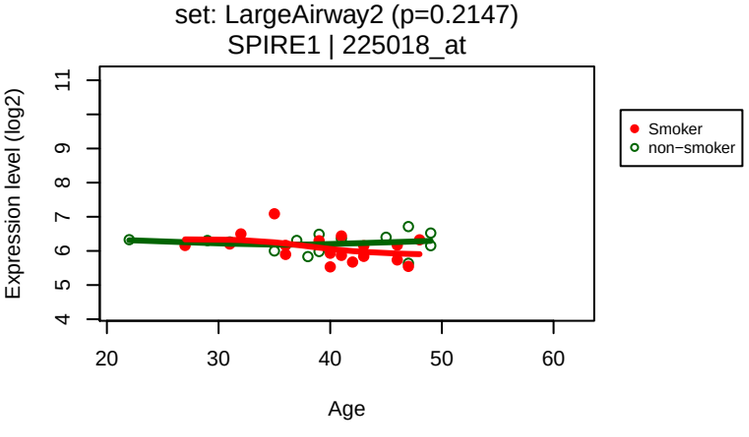

Gene expression ratio (smokers/nonsmokers)
gene: SPIRE1 | 225018_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2131
p=0.0324
p=0.0022
p=0.1422
p=0.2147
p=0.0039
p=0.9139
p=0.2816
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

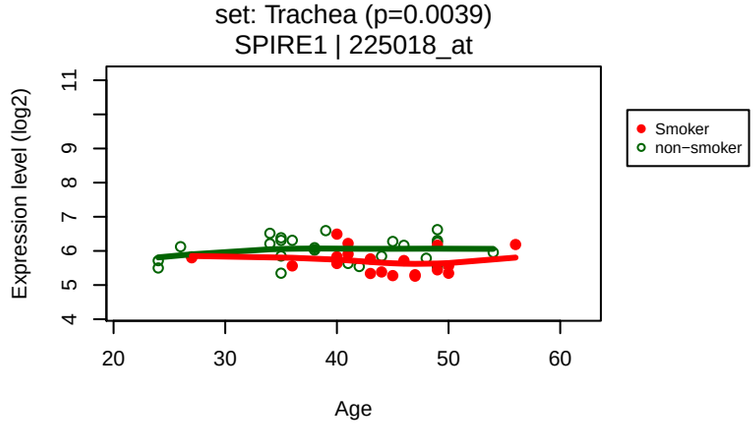

Gene expression ratio in Males
gene: SPIRE1 | 225018_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0624
p=0.8454
p=0.0266
p=0.1843
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPIRE1 | 225018_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.413
p=0.0425
p=0.0378
p=0.571
SmallAirway2
LargeAirway2
Trachea
Macrophage2

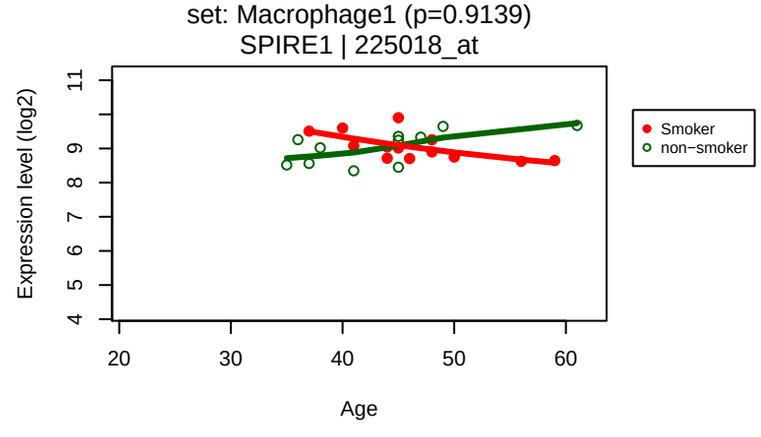

set: SmallAirway1 (p=0.2131)
SPIRE1 | 225018_at
Expression level (log2)
Age
Smoker
non−smoker

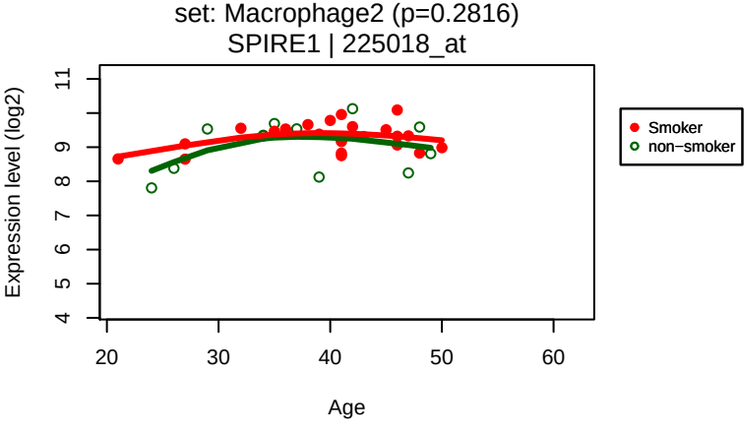

set: SmallAirway2 (p=0.0324)
SPIRE1 | 225018_at
Expression level (log2)
Age
Smoker
non−smoker

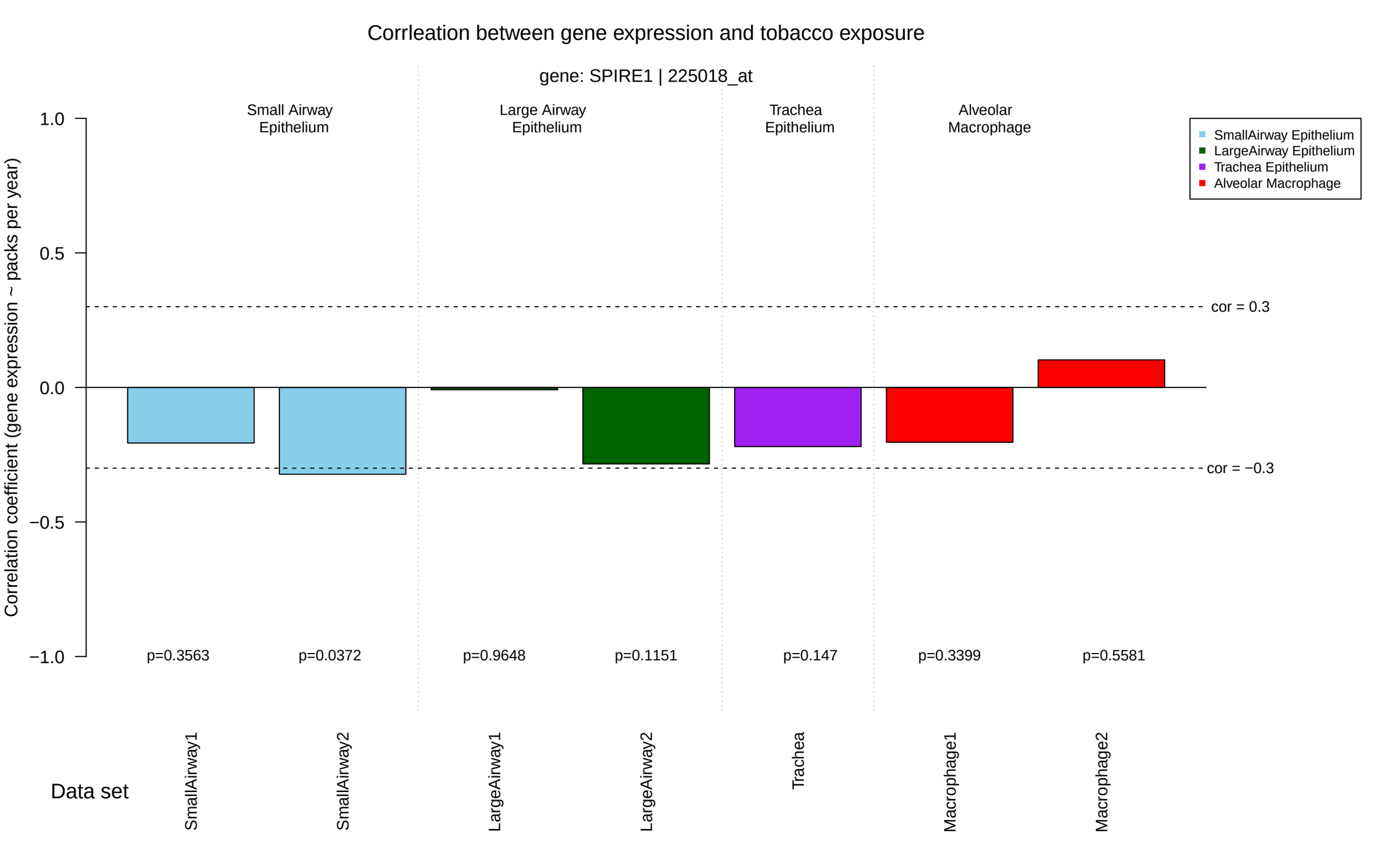

set: LargeAirway1 (p=0.1422)
SPIRE1 | 225018_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2147)
SPIRE1 | 225018_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0039)
SPIRE1 | 225018_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9139)
SPIRE1 | 225018_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2816)
SPIRE1 | 225018_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPIRE1 | 225018_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3563
p=0.0372
p=0.9648
p=0.1151
p=0.147
p=0.3399
p=0.5581
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2